

Deep Learning and its Applications to Signal and Image Processing and Analysis

361.2.1120

Final Project

Submission Date: 27.06.2025

1 Introduction

For the final project, you will choose one of three Kaggle challenges. Each challenge focuses on a different type of image processing task using deep neural networks (DNNs). For your selected task, you are required to compare two models: the first should be either a baseline model taken from a Kaggle submission or a vanilla implementation—that is, a standard, unmodified version of a commonly used architecture. The second must be your own original model or a meaningful variation of the first. Full details on the report structure are provided below.

Note: We are aware that you are working with limited computational resources, so we do not expect state-of-the-art results. However, if your model performs significantly poorly, it may result in a reduction of points.

2 Project Descriptions

Choose one of the following projects. Each challenge focuses on a different image processing task and is evaluated using a specific performance metric. Click the project titles to view their corresponding Kaggle pages and understand the task objectives and evaluation criteria in detail. To ensure the reliability of your results, **repeat the training and evaluation process three times** using different random seeds, and **report the mean and standard deviation** of your main evaluation metric(s) **each time** you required.

Note: You can sign in to the Kaggle website using your student email address. The datasets can either be downloaded manually or programmatically using the code snippet provided on each competition's webpage (typically under the "Data" tab).

2.1 Global Wheat Detection

Global Wheat Detection — Detect wheat heads in outdoor images using a robust and generalized model that performs well across diverse regions, genotypes, and imaging conditions. The evaluation metrics you need to use are Intersection over Union (IoU) and precision, as detailed on the competition webpage.

2.2 Super-Resolution Image Enhancement

DIV2K Dataset — Enhance the resolution of low-quality images by training a model on **4× downsampled** versions of high-resolution images, with the goal of accurately restoring fine

details and texture. The evaluation metrics you need to use are PSNR, SSIM and FID.

2.3 I'm Something of a Painter Myself

I'm Something of a Painter Myself — Generate realistic Monet-style paintings from photographs using generative models such as GANs, in an unpaired image-to-image translation setting. The evaluation metric you need to use is MiFID, as detailed on the competition webpage.

3 The Report

3.1 Title

The final report should contain:

1. Course name and number.
2. Final Project.
3. Names and student IDs of the submitters.
4. A link to a GitHub repository containing your code, along with a README or Markdown file that briefly presents the project. (The link can also be shared as part of a LinkedIn post!)

3.2 Abstract

Include a one-paragraph summary:

1. The challenge you chose.
2. The two models you used, with a brief description of each.
3. A summary of your conclusions.

3.3 Introduction, Objective, and Data Description

1. Clearly state the challenge.
2. Describe the dataset, including exploratory data analysis and visualizations.
3. Briefly explain the evaluation metrics used.

3.4 Model 1

For this section, you can choose one of two options:

1. A previously submitted Kaggle model with its weights.
2. Your own vanilla implementation — a basic architecture for solving the challenge.

3.4.1 Option 1: Baseline Model (Kaggle)

In this option, you are required to:

1. Provide a clear explanation of the model architecture.
2. Include a simplified diagram illustrating the model structure.
3. Describe the data preprocessing steps and include visualizations if applicable.
4. Present test results, including both the quantitative metrics and qualitative examples showing both successful and failed predictions.

3.4.2 Option 2: Your Own Implementation (Vanilla)

In this option, you are required to:

1. Provide a clear explanation of your model architecture.
2. Include a simplified diagram illustrating the model structure.
3. Describe the data preprocessing steps, including visualizations if applicable.
4. Specify the hyperparameters used and the loss function applied.
5. Include a plot of the training loss over epochs to demonstrate convergence behavior.
6. Present test results, including both quantitative metrics and qualitative examples of both positive and negative performance.

3.5 Model 2

This model should be your own original architecture or a meaningful variation of the first model.

1. Provide a clear explanation of your model architecture or the modifications made to an existing one.
2. Include a simplified diagram illustrating the model structure.
3. Describe the data preprocessing steps, including visualizations if applicable.
4. Describe the training regime, specify hyperparameters used and the loss function applied.
5. Include a plot of the training loss over epochs to demonstrate convergence behavior.
6. Present test results, including both the quantitative metrics and qualitative examples of both positive and negative performance.

3.6 Results

1. Show four examples of images:
 - One where both models performed well.
 - One where both models performed poorly.
 - One where Model 1 performed well and Model 2 did not (if none, please state).
 - One where Model 2 performed well and Model 1 did not (if none, please state).
2. Compare the performance of both models (both quantitative and qualitative). Clearly state whether your variation outperformed the baseline model, and explain the reasons behind the outcome.

3.7 Ablation Studies

Choose one meaningful component of your Model 2 (e.g., loss function, regularization technique, data augmentation strategy) and evaluate its impact on model performance.

1. Describe the ablation setup, specifying the component you varied and the motivation behind it. Include visualizations if applicable (e.g., for different augmentation methods).
2. Plot and compare the training loss curves for the original and ablated versions of the model. Briefly describe the differences observed.
3. Report quantitative performance metrics (e.g., accuracy, loss) for each version. Present the results in a clear format, such as a table.
4. Provide qualitative examples (e.g., prediction visualizations) to illustrate how the ablation affected performance. Was there an image where the ablated model performed better? Can you explain why?
5. Summarize the insights gained. What does this experiment reveal about the role of the ablated component in your model's performance?

3.8 Conclusions and Summary

Summarize key findings and overall conclusions.

3.9 References

List all references cited in your report.

4 Submission Instructions

- Work in pairs; only one submission per team is required.
- Submit the report as a PDF file.

Document Instructions

- Reports may be written in English or Hebrew.
- Use the section titles provided in this document.
- All images must be explained and clearly visible.
- Ensure the document is organized and easy to read.

Code Instructions

- Upload your code to a public GitHub repository.
- Include a clear and concise README file with instructions on how to run the code.
- Ensure the code is modular, well-organized, and reusable.
- Begin each code file with the full names and student IDs of all team members.
- Use meaningful variable and function names, and include comments where necessary.

- Use only relative paths for file access—avoid hardcoded absolute paths.
- The code must run from start to finish without errors.